

Buzzard cluster mock data vector with optical-selection effects: forward model, analytical and empirical $\mathcal{B}_{\text{sel}}$ closures

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2026-05-28

Abstract

This document is the technical reference for `MockDataVector.ipynb`: a Buzzard-halo DES Y3-like cluster mock that layers the Costanzi+2026 (C26) optical selection systematics on top of the raw simulation outputs, on the DES Y1 cluster binning ($\lambda \in \{20, 30, 45, 60, \infty\}$, $z \in \{[0.20, 0.33), [0.37, 0.50), [0.50, 0.65)\}$, which are the standard DES Y1 redshift edges with the upper edge of z_0 pulled in from $0.35 \rightarrow 0.33$ and the lower edge of z_1 pushed out from $0.35 \rightarrow 0.37$ to skip the Buzzard $0.33 \leq z < 0.37$ box-junction seam). The data vector is constructed along *two parallel routes* that share the same Costanzi+2019 (C19) forward model upstream and the same lensing geometry: an **analytical** closure based on the C26 Appendix C Eq. C1 fitting function applied to the C19-binned stack (cheap enough for an MCMC chain), and an **empirical** ratio that stacks on the catalog redMaPPer outputs (`LAMBDA.CHISQ`, `Z_LAMBDA`) and divides by a $(\log M, z^{\text{tr}})$ -matched random reference (Wu+2022, Costanzi+2026). The notebook also produces a binned number-count vector $N(\lambda^{\text{ob}}, z^{\text{ob}})$. Member-galaxy contamination (boost factor) is injected from the published DES Y1 measurements (McClintock+2019), so the saved final shear $\gamma_t^{\text{mock, obs}} = \Delta \Sigma_{\text{obs, C1}} / B_{\text{Y1}} \cdot \Sigma_{\text{crit}}^{-1}$ mimics the raw uncorrected DES Y1 data vector. All outputs are saved to `output/MockDataVector.npz`; the headline plots embedded in this document live in `output/figs/`.

Contents

1 Pipeline topology	1
2 Shared conventions	2
3 The intrinsic mass–richness relation (C26 Eq. 15)	2
4 The projection kernel (C19 Eq. 6)	3
5 Output 1 — $N(\lambda^{\text{ob}}, z^{\text{ob}})$ number counts	5
6 Lensing data vector — ANALYTICAL route (Matteo / C26 Eq. C1)	5
6.1 Boost factor: member-galaxy contamination $B(R)$	6
7 Lensing data vector — EMPIRICAL route (Heidi / mass-matched ratio)	7
7.1 Selection-bias ratio $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ with bootstrap bands	8
7.2 Null test: replace λ^{RM} with the C19 sampler	9
7.3 Tangential shear: redMaPPer-selected vs mass-matched random	9
8 Output archive	10
8.1 Units and the h -convention	11

9 Known simplifications	12
10 File map	12
11 References	12

1 Pipeline topology

The notebook executes the following stages in order:

```
halo_load → photoz_proxy → sample_lambda_true → sample_lambda_obs → lensing_inputs
→ number_counts → analytical_C1 → boost_factor → empirical_mass_match →
jackknife_covariance → save_npz
```

The first four stages are the C19 forward model: select host halos above $\log_{10} M_{\text{vir}} = 13$ in $z \in [0.20, 0.65]$ (with the Buzzard $0.33 \leq z < 0.37$ box seam excised), add a DES Y3-like Gaussian photo- z scatter to produce z^{ob} , and draw λ^{tr} and λ^{ob} from the C26/C19 kernels. `lensing_inputs` centralises the per-halo profiles and $\Sigma_{\text{crit}}^{-1}$ table, plus the catalog redMaPPer columns (`LAMBDA_CHISQ`, `Z_LAMBDA`) used by the empirical route. The two lensing-data-vector stages are independent and produce their outputs on the same DES Y1 binning so the user can compare them side-by-side. After implementing the boost factor and calculating the covariance matrices, the final stage persists everything to `output/MockDataVector.npz`.

2 Shared conventions

Cosmology. FlatLambdaCDM($H_0=70$, $\Omega_m=0.286$, $\Omega_b=0.046$, $T_{\text{cmb}}=2.725$) via `astropy.cosmology`, matching the Buzzard simulation cosmology used by the upstream halo catalogue.

Units. Per-halo Σ , $\Delta\Sigma$ are stored in physical $M_{\odot}/\text{pc}^2$; radii `rp_phys_mpc` (from `src/radial_bins_phys_mpc.py`) are in physical Mpc, with $N_R = 15$ log-spaced bins between 0.0323 and 30 Mpc. Halo masses M_{vir} are in $h^{-1} M_{\odot}$.

Halo selection. Mirrors `0-MakeMock.ipynb`: `pid == -1` (host halos only), $0 \leq \cos i \leq 1$ (measured), drop the $0.33 \leq z < 0.37$ simulation-box seam, then require $0.20 \leq z^{\text{tr}} \leq 0.65$ and $\log_{10} M_{\text{vir}} \geq 13$. Yields $\sim 5.83 \times 10^5$ halos.

DES Y1 binning (used everywhere). Both abundance and lensing bins follow the DES Y1 cluster edges (Costanzi+2019, McClintock+2019), with the central two redshift edges shifted inward by ± 0.02 to skip the Buzzard box-junction seam at $0.33 \leq z < 0.37$:

$$\text{LBDBINS} = \{20, 30, 45, 60, 500\}, \quad \text{ZMIN_LIST} = \{0.20, 0.37, 0.50\}, \quad \text{ZMAX_LIST} = \{0.33, 0.50, 0.65\}.$$

The 500 caps the open $[60, \infty)$ richness range. There are $N_{\lambda} \times N_z = 4 \times 3 = 12$ bins. The seam volume is $\sim 5\%$ of z_0 and $\sim 13\%$ of z_1 on the published Y1 edges; pulling the bin edges inward eliminates the bias on $N(\lambda^{\text{ob}}, z^{\text{ob}})$ at the cost of a small mismatch with the Y1-published bin ranges (negligible on B_{sel} , B_{data} , and the shear data vector since these are intensive quantities).

Photo- z proxy. Per-halo Gaussian

$$z^{\text{ob}} = z^{\text{tr}} + \sigma_z(1 + z^{\text{tr}})\mathcal{N}(0, 1), \quad \sigma_z = 0.01, \quad (1)$$

clipped at $z^{\text{ob}} \geq 0.02$. Stand-in for the DES Y3 redshift kernel of Myles+2021; captures the dominant scatter but not the tails.

Random seeds. A single `numpy.random.default_rng(seed=42)` is shared across the forward model. The MC-vs-PDF sanity histogram uses `seed=123`; the §8.1 bootstrap uses `seed=0`.

3 The intrinsic mass–richness relation (C26 Eq. 15)

`Path` `src/costanzi_selection.py (l_sat, l_tr, sample_lambda_true)`.

The intrinsic richness λ^{tr} of a halo (the true number of red-sequence satellite galaxies hosted by the halo, plus the central galaxy) is modelled as a halo central plus a Poisson satellite count. The mean satellite count $\bar{\lambda}_{\text{sat}}(M, z)$ follows C26 Eq. 15,

$$\bar{\lambda}_{\text{sat}}(M, z) = \left(\frac{M - M_{\text{min}}}{M_{\text{pivot}}} \right)^\alpha \left(\frac{1 + z}{1 + z_{\text{piv}}} \right)^\epsilon, \quad (2)$$

clipped to zero for $M < M_{\text{min}}$. The mean intrinsic richness adds the central galaxy when the halo is above threshold:

$$\langle \lambda^{\text{tr}} | M, z \rangle = \begin{cases} 1 + \bar{\lambda}_{\text{sat}}(M, z), & M \geq M_{\text{min}}, \\ 0, & M < M_{\text{min}}. \end{cases} \quad (3)$$

The DES Y1 NC+3×2pt best-fit constants, hard-coded in `src/costanzi_selection.py`:

$$\log_{10} M_{\text{min}} = 11.385, \quad \alpha = 0.859, \quad \log_{10} M_1 = 12.696, \quad \sigma_{\text{intr}} = 0.181, \quad \epsilon = 0.284, \quad z_{\text{piv}} = 0.4544,$$

with $M_{\text{pivot}} = M_1 - M_{\text{min}}$.

The per-halo λ^{tr} value is drawn (C26 Sec. III) by (i) broadening the Poisson rate $\bar{\lambda}_{\text{sat}}$ with a multiplicative log-normal of width σ_{intr} , (ii) drawing an integer satellite count, and (iii) adding 1 for the central galaxy if the halo is above M_{min} :

$$\tilde{\lambda}_{\text{sat}} = \bar{\lambda}_{\text{sat}}(M, z) \exp\left[\sigma_{\text{intr}} \mathcal{N}(0, 1) - \frac{1}{2}\sigma_{\text{intr}}^2\right], \quad N_{\text{sat}} \sim \text{Poisson}(\tilde{\lambda}_{\text{sat}}), \quad \lambda^{\text{tr}} = N_{\text{cen}}(M) + N_{\text{sat}},$$

where $N_{\text{cen}}(M) = 1$ if $M \geq M_{\text{min}}$ and 0 otherwise.

4 The projection kernel (C19 Eq. 6)

`Path` `src/costanzi_selection.py (projection_params, sample_lambda_obs)`. C19 Eq. 6 is a Gaussian-core + exponentially-modified-Gaussian (EMG) mixture:

$$P(\lambda^{\text{ob}} | \lambda^{\text{tr}}, z) = (1 - f_{\text{prj}}) \mathcal{N}(\mu, \sigma) + f_{\text{prj}} \text{EMG}(\mu, \sigma, \tau). \quad (4)$$

The four parameters $\{\mu, \sigma, \tau, f_{\text{prj}}\}$ depend on λ^{tr} with z -dependent coefficients calibrated from SDSS redMaPPer (C19 Appendix A, Eq. A8):

$$\begin{aligned} \mu(\lambda^{\text{tr}}, z) &= a_\mu(z) + b_\mu(z) \lambda^{\text{tr}}, & \sigma(\lambda^{\text{tr}}, z) &= b_\sigma(z) (\lambda^{\text{tr}})^{a_\sigma(z)}, \\ \tau(\lambda^{\text{tr}}, z) &= b_\tau(z) (\lambda^{\text{tr}})^{-a_\tau(z)}, & f_{\text{prj}}(\lambda^{\text{tr}}, z) &= b_{f_{\text{prj}}}(z) (1 + e^{-\lambda^{\text{tr}}})^{-a_{f_{\text{prj}}}(z)}. \end{aligned} \quad (5)$$

The 15 rows of `data/prj_params_DESY3_lss_lin_dep_getdist_v1.txt` are interpreted as posterior samples and collapsed to the posterior mean via `load_prj_posterior_mean`.

The mixture is sampled directly using the $\text{EMG} = \mathcal{N}(\mu, \sigma) + \text{Exp}(1/\tau)$ representation, avoiding a closed-form EMG density evaluation in the sampling path. The closed form is used only for the sanity overlay below.

Sanity check 1 — MC draws vs analytical kernel. Histogram repeated `sample_lambda_obs` draws at fixed (λ^{tr}, z) and overlay the closed-form C19 PDF; they should agree within Poisson noise.

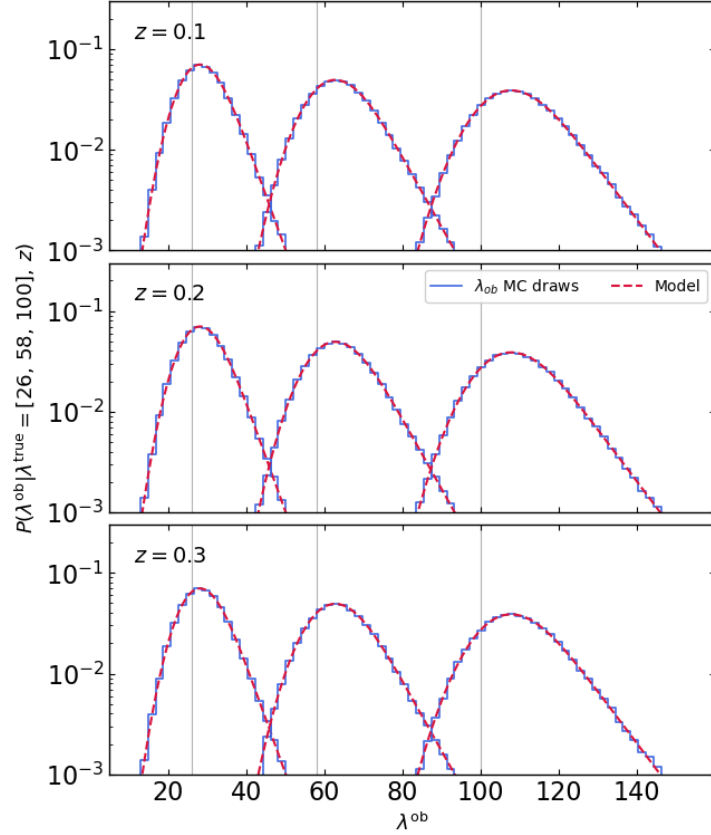


Figure 1: MC draws of λ^{ob} from `sample_lambda_obs` (blue histograms) overlaid with the closed-form PDF of Eq. 4 (red dashed) at three reference redshifts and three intrinsic richnesses. The two agree within Poisson noise across the bulk and the tail.

Sanity check 2 — Gaussian core + EMG tail decomposition. Same kernel as Eq. 4, split into the two physical components. At large λ^{tr} the EMG tail dominates the high- λ^{ob} side and drives the projection bias; this is what the analytical $\mathcal{B}_{\text{sel}, \text{C1}}$ of §6 fits.

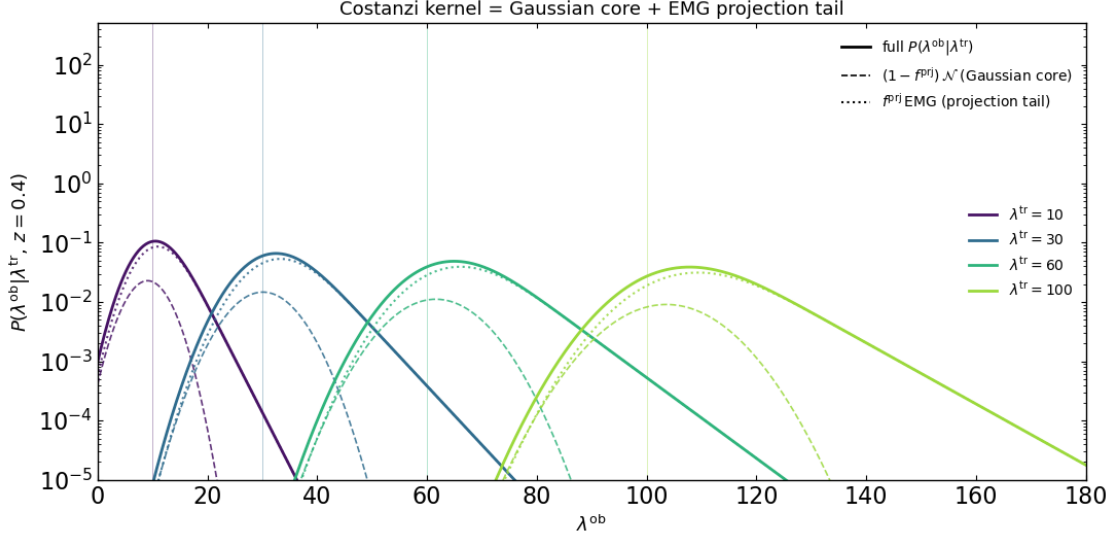


Figure 2: Decomposition of the kernel of Eq. 4 into its $(1 - f_{\text{prj}})\mathcal{N}$ Gaussian core (dashed) and f_{prj} EMG projection tail (dotted) at $z = 0.4$ for four intrinsic richnesses $\lambda^{\text{tr}} \in \{10, 30, 60, 100\}$. The EMG tail rises with λ^{tr} and drives the optical-selection bias on the lensing data vector.

5 Output 1 — $N(\lambda^{\text{ob}}, z^{\text{ob}})$ number counts

We obtain $N(\lambda^{\text{ob}}, z^{\text{ob}})$ by counting the mock halos that fall into each $(\lambda^{\text{ob}}, z^{\text{ob}})$ bin on the DES Y1 grid. Both axes are forward-model outputs rather than catalog redMaPPer columns: the C19 sampler depends only on (M, z) and carries no per-halo LSS information, which matches the inputs of the analytical $\mathcal{B}_{\text{sel}, \text{C1}}(\lambda^{\text{ob}}, z^{\text{ob}})$ of §6 and keeps the abundance and lensing sides of the data vector in the same C26/C19 model world. $N(\lambda^{\text{ob}}, z^{\text{ob}})$ is therefore the correct entry for a C26 likelihood integrating Eq. 2 into Eq. 4.

We also compute the covariance matrix for the number counts by only considering the Poisson noise of the counts in each bin, which is a diagonal matrix with

$$\text{Cov}[N(\lambda^{\text{ob}}, z^{\text{ob}})]_{ii} = \sqrt{N(\lambda^{\text{ob}}, z^{\text{ob}})_i}. \quad (6)$$

6 Lensing data vector — ANALYTICAL route (Matteo / C26 Eq. C1)

Binning quantity. The C19 forward-model $(\lambda^{\text{ob}}, z^{\text{ob}})$. Each λ^{ob} is an independent draw from the kernel of Eq. 4 conditioned only on (λ^{tr}, z) : the kernel encodes the LSS-driven projection statistics through its EMG tail at the population level, but per-halo λ^{ob} values are not correlated with the actual line-of-sight structure around each halo and therefore carry no information correlated with the per-halo lensing signal.

Selection bias. redMaPPer counts red-sequence galaxies inside a projected aperture with no line-of-sight resolution, so projected LSS preferentially boosts λ^{RM} for halos in over-dense filaments and lifts the stacked $\Delta\Sigma(R)$ above the mass-only expectation. C26 Appendix C captures this lift with a 5-parameter fitting function (no integration); Matteo (priv. comm.) reports the same form fits the bias on $\Delta\Sigma$:

$$\mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}(R) = A \left(\frac{R}{R_0} \right)^\alpha \left[1 + \left(\frac{R}{R_0} \right)^\gamma \right]^{(\beta - \alpha)/\gamma} + C, \quad (7)$$

with $(A, \alpha, \beta, \gamma, C) = (0.12, 4.11, -0.18, 1.82, 1.00)$, $R_0 = R_\lambda(\lambda^{\text{ob}})(1+z^{\text{ob}})$, and $R_\lambda = (\lambda^{\text{ob}}/100)^{0.2} h^{-1} \text{ Mpc}$ (C26 Appendix B).

The C26 paper writes the published form for Σ with a fixed +1 rather than a free C ($C = 1$ limit of Eq. 7); we plot both flavours for comparison. The data vector is

$$\Delta\Sigma_{\text{obs}}(R) = \mathcal{B}_{\text{sel},C1}^{\Delta\Sigma}(R) \Delta\Sigma_{\text{stack}}(R), \quad \langle\gamma_t(R)\rangle = \Delta\Sigma_{\text{obs}}(R) \Sigma_{\text{crit}}^{-1}(\langle z^{\text{ob}} \rangle_{\text{bin}}), \quad (8)$$

with $\Delta\Sigma_{\text{stack}}$ the unweighted mean of `DeltaSigma` across the C19-binned halos in each $(\lambda^{\text{ob}}, z^{\text{ob}})$ cell.

Why a fitting function? An MCMC chain evaluates the lensing model $\sim 10^4$ – 10^6 times, so the per-step cost matters. Computing the full Costanzi-2026 prediction from scratch requires nested integrals over halo mass and redshift at every step — prohibitive at chain length. Equation (7) replaces those integrals with five constants per bin, cutting the cost from seconds to microseconds without changing the answer.

Selection-bias ratio $\mathcal{B}_{\text{sel},C1}^{\Delta\Sigma}(R)$ The fitting function evaluated at the bin-mean $(\langle\lambda^{\text{ob}}\rangle, \langle z^{\text{ob}}\rangle)$ for every $(\lambda^{\text{ob}}, z^{\text{ob}})$ cell, with the published Σ -flavour overlay.

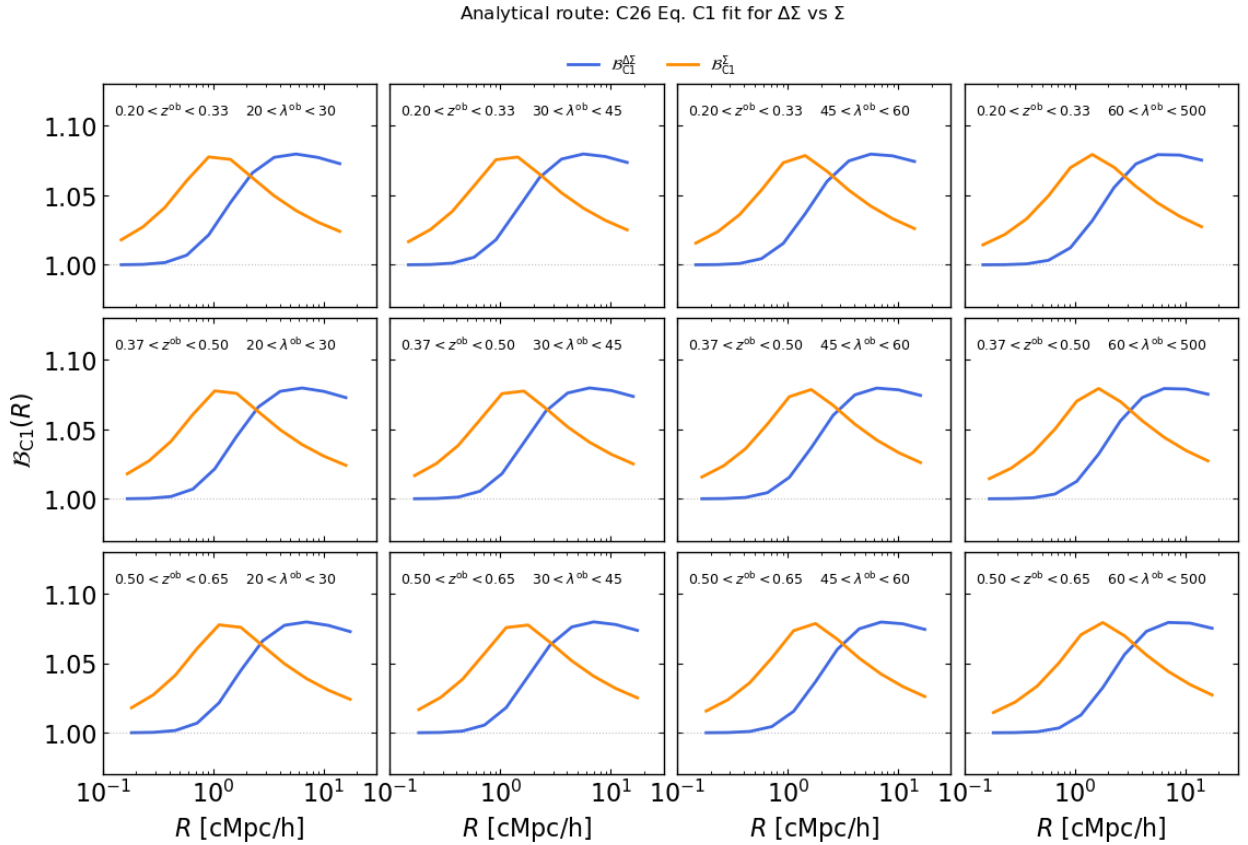


Figure 3: Analytical route: C26 Eq. C1 selection-bias ratio per $(\lambda^{\text{ob}}, z^{\text{ob}})$ bin on the DES Y1 grid. Blue: $\mathcal{B}_{\text{sel},C1}^{\Delta\Sigma}$ (used in the data vector). Orange: $\mathcal{B}_{\text{sel},C1}^{\Sigma}$ overlay. x -axis is comoving R in $h^{-1} \text{ Mpc}$ to match the C26 convention; y -axis is the bias factor.

6.1 Boost factor: member-galaxy contamination $B(R)$

A real DES Y1 measurement of $\Delta\Sigma$ is diluted by cluster-member galaxies that survive photo- z cuts and end up in the lensing source catalog. These members carry no shear signal and reduce the measured

profile by the boost factor $B(R) \equiv 1/(1 - f_{\text{cl}}) \geq 1$ (McClintock+2019 §5.3.1).

$$B(R; r_s, b_0) = 1 + b_0 \frac{1 - f(x)}{x^2 - 1}, \quad x = R/r_s,$$

$$f(x) = \begin{cases} \frac{\arctan \sqrt{x^2 - 1}}{\sqrt{x^2 - 1}}, & x > 1 \\ \frac{\operatorname{arctanh} \sqrt{1 - x^2}}{\sqrt{1 - x^2}}, & x < 1 \end{cases} \quad (9)$$

To make the saved data vector mock the *raw uncorrected* DES Y1 measurement, both lensing routes have their final shear stack divided by $B(R)$:

$$\langle \gamma_t \rangle^{\text{mock, obs}} = \frac{\langle \gamma_t \rangle^{\text{obs}}}{B_{\text{Y1 data}}(R)} \quad (10)$$

Where $B_{\text{Y1 data}}(R)$ comes from. The DES Y1 cluster analysis released measured boost-factor profiles per richness/redshift bin, located at `data/boost_factor/profiles/ (full-unblind-v2-mcal-zmix_y1clust_1{1}-z{`. The files are tabulated on an 11-point physical-Mpc R grid (0.166 to 15.80 Mpc); we adopt this grid as the global radial axis for the saved data vector. The 7 cosmology-side richness bins 10..16 include $\lambda < 20$ ranges; only 13..16 match our 4 lensing bins $\lambda \in \{[20, 30), [30, 45), [45, 60), [60, \infty)\}$. The parametric McClintock+2019 Table B1 fit is also computed (cell 42) for sanity but is not used in the saved output.

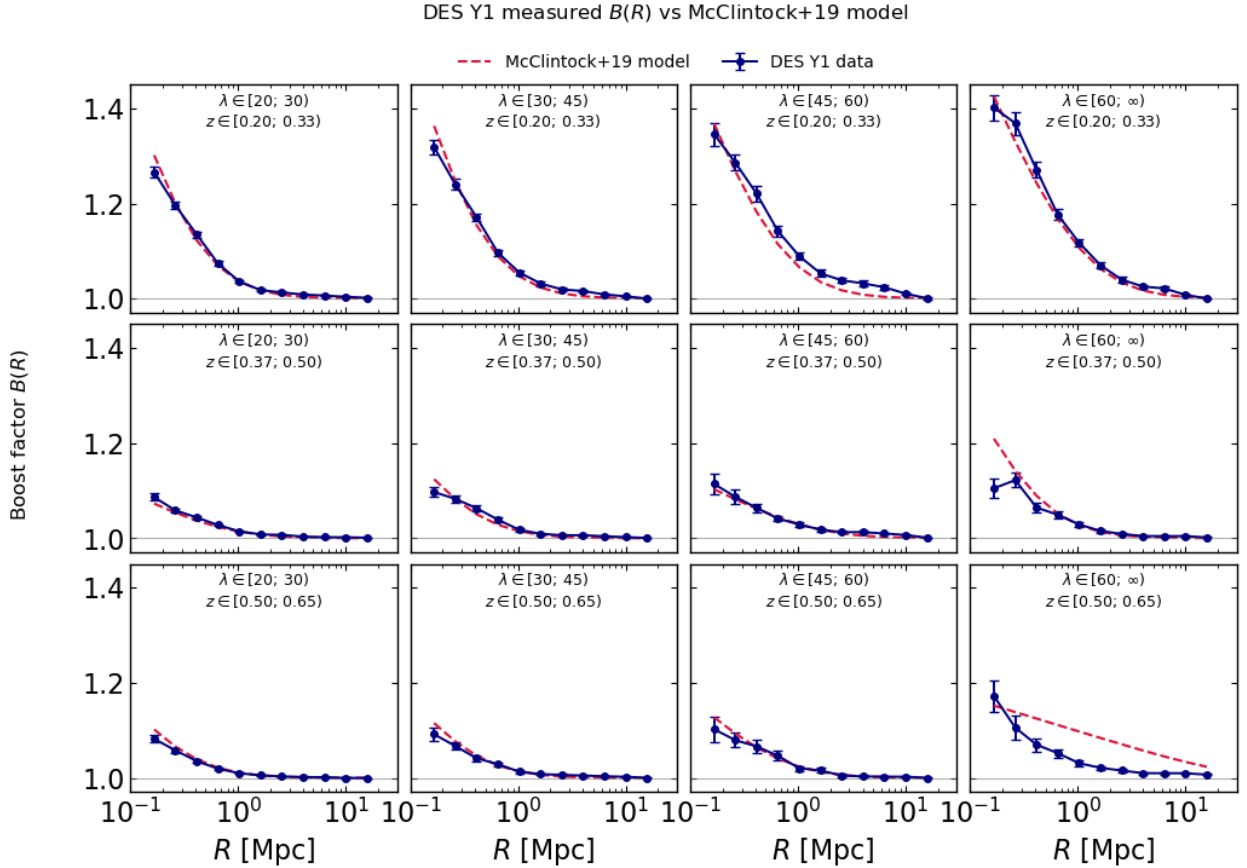


Figure 4: Measured DES Y1 boost factor $B(R)$ per $(\lambda^{\text{RM}}, z^{\text{RM}})$ bin with reported errors. Small- R amplitude rises with richness and falls with redshift, as expected for a red-sequence-driven member contamination.

We

7 Lensing data vector — EMPIRICAL route (Heidi / mass-matched ratio)

Binning quantity. The catalog redMaPPer outputs $(\lambda^{\text{RM}}, z^{\text{RM}})$, i.e. `LAMBDA_CHISQ` and `Z_LAMBDA`. These come from running the redMaPPer cluster finder on the Buzzard photometric catalog and *do* carry per-halo LSS information.

Selection bias. Built directly from the catalog as the ratio of the redMaPPer-selected stack to a $(\log M, z^{\text{tr}})$ -matched random reference, following Wu+2022 method iii (`src/stacked_profile_weighted_by_mass_redsh`). For each $(\lambda^{\text{RM}}, z^{\text{RM}})$ bin:

$$\langle \Delta \Sigma(R) \rangle_{\text{rnd}} = \frac{\sum_c w_c \langle \Delta \Sigma(R) | c \rangle_{\text{full mock}}}{\sum_c w_c}, \quad w_c = N_c^{\text{sel}}, \quad (11)$$

summed over $(\log_{10} M, z^{\text{tr}})$ cells of width $(\Delta \log M, \Delta z) = (0.1, 0.05)$. The cell averages are taken over the *full* mock, so the reference resamples halos that share the selected sample’s mass and redshift but were never preferentially picked for their environment. The empirical bias is

$$\mathcal{B}_{\text{sel}}^{\text{emp}}(R) = \frac{\langle \Delta \Sigma(R) \rangle_{\text{sel}}}{\langle \Delta \Sigma(R) \rangle_{\text{rnd}}}.$$

No fitting function, no $R \rightarrow 0$ extrapolation, both numerator and denominator come straight from `DeltaSigma`.

Bootstrap. The reported $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ is the **bootstrap mean** over $N_{\text{boot}} = 50$ resamples of halos within each bin (with mass-matched reference rebuilt at every step), capturing the statistical uncertainty driven by finite halo count. The shaded bands in the panel grids are the bootstrap standard deviation; the saved file keeps only the bootstrap mean.

7.1 Selection-bias ratio $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ with bootstrap bands

We evaluate $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ on the redMaPPer-binned $(\lambda^{\text{RM}}, z^{\text{RM}})$ grid using Eq. (11) and report the bootstrap mean and 1σ band per bin. The expected signature is the bell-shaped excess peaking near $R \sim 1$ pMpc that Wu+2022 (Fig. 2) identify as the LSS-environment-driven selection bias: at smaller scales the one-halo term dominates and the ratio approaches unity, while at larger scales the projected over-density that boosted λ^{RM} falls outside the lensing aperture and the ratio relaxes back toward one.

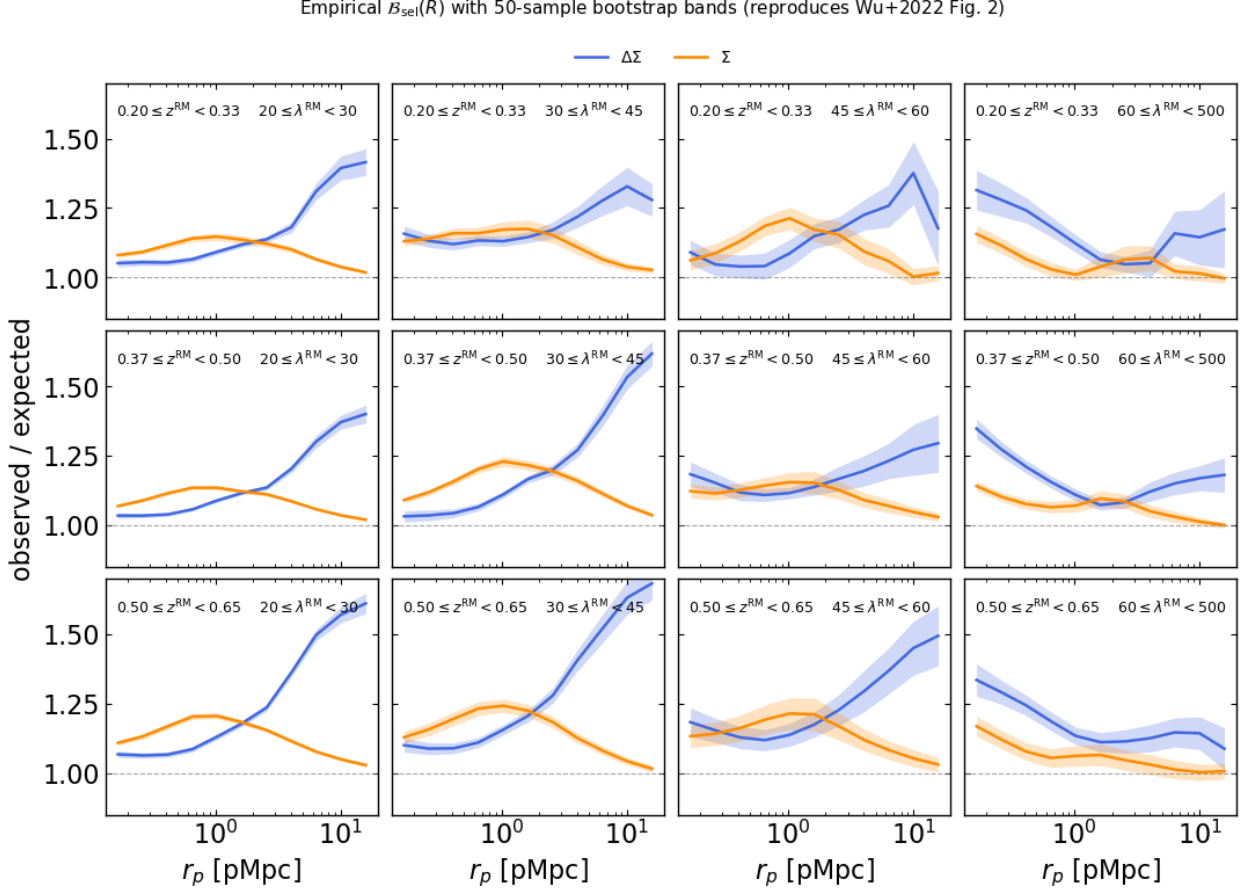


Figure 5: Empirical route: $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ per $(\lambda^{\text{RM}}, z^{\text{RM}})$ bin on the DES Y1 grid, with $N_{\text{boot}} = 50$ bootstrap bands. Blue: $\Delta\Sigma$ ratio. Orange: Σ ratio (overlay). Reproduces Wu+2022 Fig. 2: the bell-shaped peak around $R \sim 1$ pMpc is the LSS-environment-driven selection bias signal.

7.2 Null test: replace λ^{RM} with the C19 sampler

If we re-bin on the C19 forward-model $(\lambda^{\text{ob}}, z^{\text{ob}})$ instead of the catalog $(\lambda^{\text{RM}}, z^{\text{RM}})$, the same mass-matched-random pipeline should return unity at every R , because the C19 sampler is by construction blind to per-halo LSS. Any residual deviation from 1 is the bias of the matching procedure itself, not a physical signal.

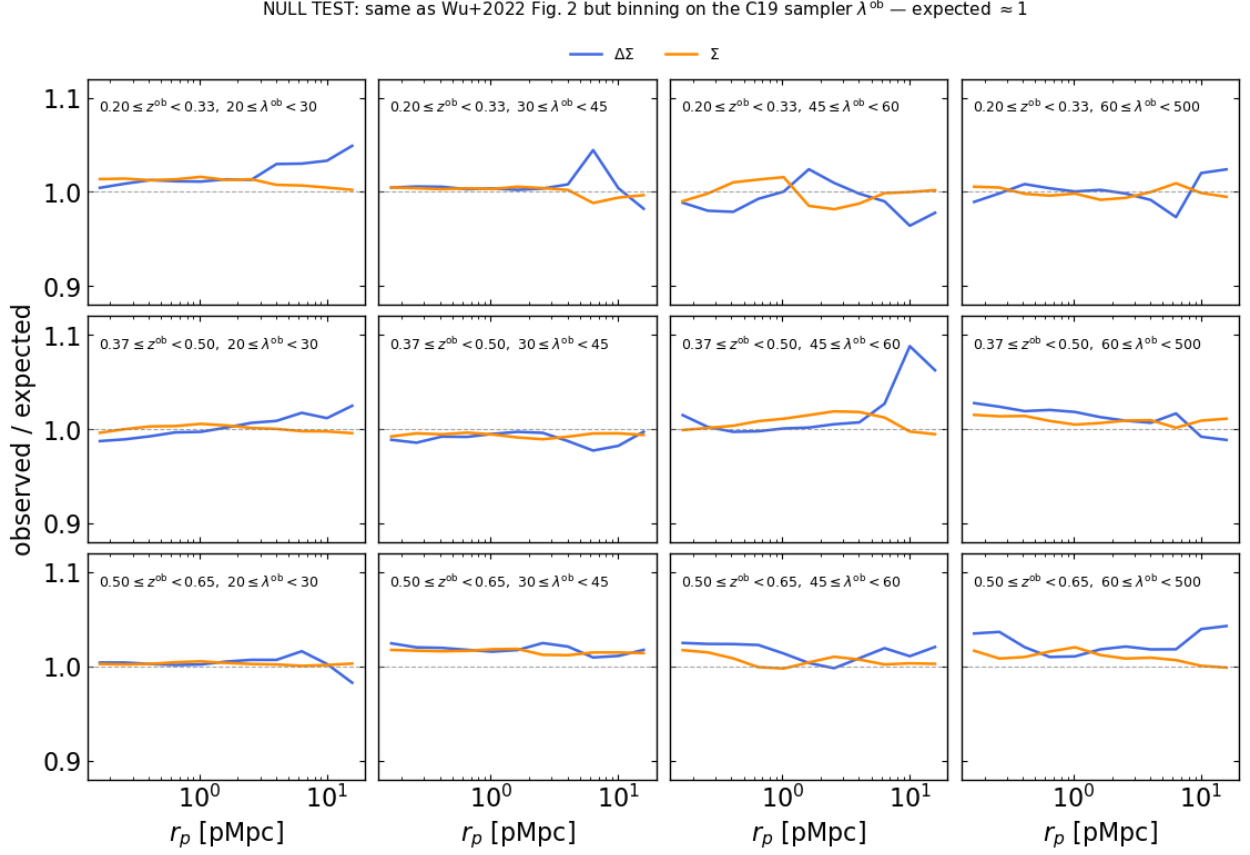


Figure 6: Null test: same Wu+22 panel grid but binning on the C19 sampler ($\lambda^{\text{ob}}, z^{\text{ob}}$) rather than the catalog ($\lambda^{\text{RM}}, z^{\text{RM}}$). The bell collapses to ≈ 1 across all bins, confirming the $R \sim 1$ pMpc peak in Fig. 5 is genuinely LSS-driven.

7.3 Tangential shear: redMaPPer-selected vs mass-matched random

The bias ratio of §7.1 is most directly seen by comparing the two tangential-shear stacks that enter its numerator and denominator. The mass-matched random reference is the γ_t a hypothetical unbiased selector would measure on halos of the same $(\log M, z^{\text{tr}})$ distribution; the redMaPPer-selected stack is the γ_t an actual cluster finder produces. Their offset is the observable counterpart of $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ and the quantity an MCMC must reproduce.

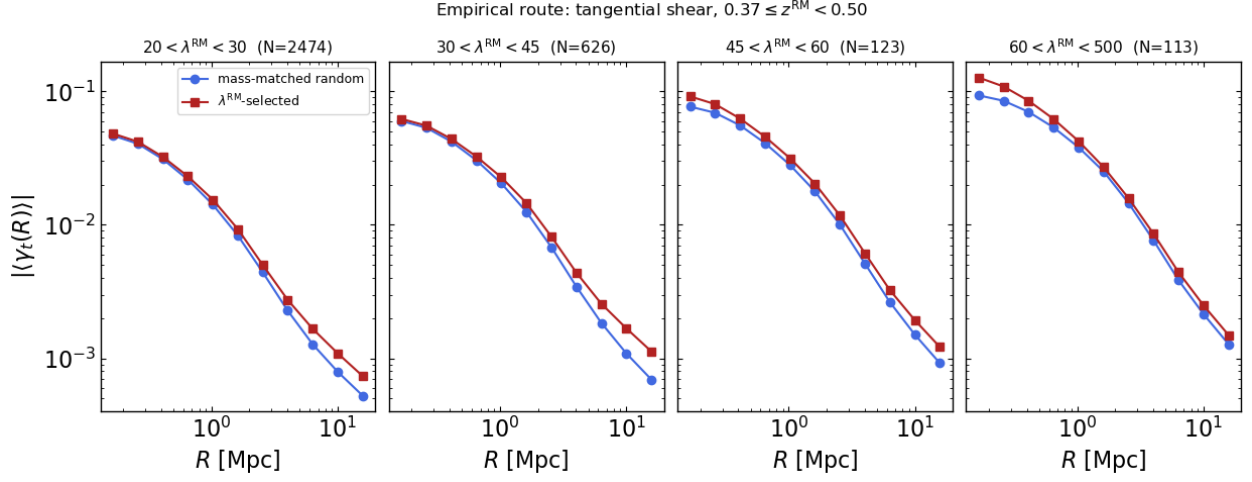


Figure 7: Empirical route: per-bin tangential shear in the middle redshift slice $z^{\text{RM}} \in [0.37, 0.50)$. Blue circles: γ_t from the mass-matched random reference. Red squares: γ_t from the λ^{RM} -selected stack (the observed signal). The selected curve sits above the reference by the $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ factor of Fig. 5.

8 Output archive

The final notebook cell writes both routes plus bin metadata to a single `output/MockDataVector.npz`. With $N_\lambda = 4$, $N_z = 3$, $N_R = 11$ (DES Y1 boost-factor profile grid):

Key	Shape	Content
<i>Bin metadata</i>		
<code>radii_phys_mpc</code>	$(N_R,)$	11-pt DES Y1 boost grid (physical Mpc; 0.166–15.80)
<code>lambda_bins</code>	$(N_\lambda + 1,)$	DES Y1 richness edges $\{20, 30, 45, 60, 500\}$
<code>z_bin_min</code>	$(N_z,)$	lower-redshift edges
<code>z_bin_max</code>	$(N_z,)$	upper-redshift edges
<i>Lensing geometry, C19/analytical route only (§8.1)</i>		
<code>scinv_bin_C19</code>	(N_λ, N_z)	$\Sigma_{\text{crit}}^{-1}(\langle z^{\text{ob}} \rangle_{\text{bin}})$, physical $[\text{pc}^2/M_\odot]$
<code>z_rep_C19</code>	(N_λ, N_z)	bin-mean $\langle z^{\text{ob}} \rangle$ used to evaluate <code>scinv_bin_C19</code>
<code>h_buzzard</code>	scalar	Buzzard’s fixed $h = 0.70$ (DeRose+2019)
<code>scinv_bin_C19_littleh</code>	(N_λ, N_z)	$\Sigma_{\text{crit}}^{-1}$, little- h $[\text{pc}^2/(h M_\odot)] = \text{scinv_bin_C19}/h_buzzard$
<code>radii_cMpc_h_C19</code>	(N_λ, N_z, N_R)	comoving h^{-1} Mpc coordinate of <code>radii_phys_mpc</code> , per bin
<i>Number counts (DES Y1 expectation)</i>		
<code>NC</code>	(N_λ, N_z)	$N(\lambda^{\text{ob}}, z^{\text{ob}})$, Buzzard counts rescaled by $\Omega_{Y1}(z)$
<i>Analytical route, C19-binned</i>		
<code>gamma_t_stack_C19</code>	(N_λ, N_z, N_R)	stacked γ_t , no correction
<code>gamma_t_obs_C1</code>	(N_λ, N_z, N_R)	stack $\times \mathcal{B}_{\text{sel},C1}^{\Delta\Sigma}$
<code>B_sel_C1</code>	(N_λ, N_z, N_R)	analytical $\mathcal{B}_{\text{sel},C1}^{\Delta\Sigma}(R)$
<i>Empirical route, RM-binned</i>		
<code>gamma_t_stack_RM</code>	(N_λ, N_z, N_R)	$(\log M, z^{\text{tr}})$ -matched random reference γ_t
<code>gamma_t_obs_RM</code>	(N_λ, N_z, N_R)	λ^{RM} -selected stack γ_t (the observed signal)
<code>B_sel_emp</code>	(N_λ, N_z, N_R)	bootstrap-mean $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$
<i>DES Y1 boost factor + final mock-observed shears (§6.1)</i>		
<code>B_data</code>	(N_λ, N_z, N_R)	measured DES Y1 $B(R)$
<code>B_data_err</code>	(N_λ, N_z, N_R)	1- σ error per radial bin
<code>B_data_cov</code>	$(N_\lambda, N_z, N_R, N_R)$	full radial covariance matrix per bin
<code>gamma_t_mock_obs_C1</code>	(N_λ, N_z, N_R)	analytical (Matteo) shear with Y1 boost dilution
<code>gamma_t_mock_obs_RM</code>	(N_λ, N_z, N_R)	empirical (Heidi) shear with Y1 boost dilution

Table 1: Output-archive layout. Default shapes are $(4, 3)$ for the counts and $(4, 3, 15)$ for the radial vectors.

8.1 Units and the h -convention

Every `gamma_t_*` array is dimensionless ($\gamma_t = \Delta\Sigma \times \Sigma_{\text{crit}}^{-1}$) and computed self-consistently in physical units throughout, so it never needs an h -conversion and is identical regardless of which `scinv_bin_C19*` convention below it is paired with:

- **Physical** (`radii_phys_mpc`, `scinv_bin_C19`): proper Mpc and $\text{pc}^2/M_\odot$, at Buzzard’s fixed cosmology $h = 0.70$. This is an absolute-unit quantity, not a little- h -parametrized one — Buzzard’s cosmology is not a free parameter of the mock.
- **Little- h** (`radii_cMpc_h_C19`, `scinv_bin_C19_littleh`): the standard weak-lensing convention, comoving h^{-1} Mpc and $\text{pc}^2/(h M_\odot)$. `gamma_t_mock_obs_C1` / `scinv_bin_C19_littleh` recovers $\Delta\Sigma$ directly in $h M_\odot/\text{pc}^2$.

Both `*_C19` fields are valid *only* for the analytical/Matteo route (`gamma_t*_C19/gamma_t_stack_C19`) — the empirical/RM route (`gamma_t*_RM`) uses its own internal bin-mean redshift from the redMaPPer

catalog (§7) that is not currently exposed in the archive. `output/MockDataVector_targetR.npz` (a separate, secondary artifact) is not covered by either convention above and remains in its previous, unconverted physical-Mpc convention.

9 Known simplifications

- **Photo- z kernel.** The Gaussian $\sigma_z = 0.01(1 + z^{\text{tr}})$ proxy is a stand-in for the redshift- and colour-dependent kernel of Myles+2021. It captures the dominant scatter but misses the tails that drive the high- z outlier rate.
- **Boost factor uses DES Y1, not Y3, measurements.** §6.1 injects $B(R)$ from McClintock+2019 Y1 profiles, which is the closest released measurement matching our cluster binning. A Y3 boost-factor measurement is not yet released; when it is, swap the source files in `data/boost_factor/profiles/` without further code change.
- **Posterior-mean projection coefficients.** The C19 projection parameters are evaluated at the posterior mean of the upstream chain. Row-by-row covariance propagation is not implemented.
- **Single Y3 realisation.** The §8.1 empirical ratio is computed on one Y3 realisation; Wu+2022 averages 12 Y1 + 1 Y3 realisations and is correspondingly tighter.

10 File map

- `MockDataVector.ipynb` — the production notebook. Top-to-bottom execution reproduces every figure and writes the output archive.
- `src/costanzi_selection.py` — C26 Eq. 15 sampler and C19 Eq. 6 mixture sampler, plus the legacy single-scale $\mathcal{B}_{\text{sel}}$ weight.
- `src/stacked_profile_weighted_by_mass_redshift.py` — Wu+22 / Esteves (M, z)-matched random reference builder.
- `src/radial_bins_phys_mpc.py` — the 15-bin physical radial grid `rp_phys_mpc`.
- `data/prj_params_DESY3_lss_lin_dep_getdist.v1.txt` — 15 posterior samples of the 10 C19 projection coefficients.
- `src/fileLoc.py` — machine-resolved path table for the raw Buzzard halo catalogue, profile output, and β_{eff} table.
- `output/MockDataVector.npz` — saved data vector (Tab. 1).
- `output/figs/` — headline plots embedded in this PDF.
- `docs/MockDataVector.tex` — this document.

11 References

- The Buzzard mock catalogues are described in DeRose et al. 2019, ApJS 242, 8.
- The parameters for the scaling relations in Eq. 2 and projection in Eq. 5 are taken from <https://github.com/MC>
- Costanzi et al. 2026, *Forward analytical model for the optical selection bias on galaxy cluster lensing profiles*. Eqs. 15 (mass–richness), Appendix C Eq. C1 (selection-bias fit), Sec. III (Poisson–Gaussian convolution).

- Costanzi et al. 2019, arXiv:1807.07072. Eq. 6 (Gaussian + EMG mixture), Appendix A Eq. A8 (per-parameter forms).
- Wu et al. 2022, arXiv:2203.05416. Fig. 2 (selection-bias panel grid) and Appendix B (mass-redshift weighted reference).
- McClintock et al. 2019, MNRAS 482, 1352 — DES Y1 cluster binning convention used throughout.